# RANK-FOURTEEN OBSTRUCTION TO SAME-RANK EVALUATION APPROXIMATION

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> We conjecture that it is possible to take $N_m$ as the rank of $H$, i.e. that the set of continuous Hankel forms of rank $\le r$ coincides with the closure (in any reasonable topology) of the set $\{(f,g) \mapsto \sum_{j=1}^{r} c_j f(z_j)g(z_j) : c_1, \dots, c_r \in \mathbb{C}, z_1, \dots, z_r \in \Omega\}$. (For $n = 1$, this follows easily from (14.3), but we have been unable to find a proof in higher dimension.)

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Remark 14.2, p. 126.

Here $\Omega \subset \mathbb{C}^n$ is the analytic domain and $H(f,g)$ depends only on the product $fg$. Formula (14.3) describes a one-variable finite-codimension polynomial ideal as $J = \{f : f^{(\nu)}(z_j) = 0,\ 0 \le \nu < k_j,\ 1 \le j \le N\}$; coalescing point evaluations approximate these derivative evaluations.

**Answered: no.** Evidence: Sketch.

No in general: a continuous rank-fourteen Hankel form in six variables is not a pointwise limit of sums of at most fourteen point-evaluation forms.

*Proof.* Jelisiejew, *VSPs of cubic fourfolds and the Gorenstein locus of the Hilbert scheme of 14 points on* $\mathbb{A}^6$, doi:10.1016/j.laa.2018.08.002, Theorem 1.1, parts 3 and 4, gives a non-cone cubic $C = \sum_{|\alpha|=3} c_\alpha u^\alpha$ in six variables whose contraction-apolar algebra has Hilbert function $(1,6,6,1)$ and is nonsmoothable. Convert contraction to ordinary differentiation by putting $F(u) = \sum_{|\alpha|=3} c_\alpha u^\alpha/\alpha!$, where $\alpha! = \prod_{j=1}^{6} \alpha_j!$. For $p(z) = \sum_\beta p_\beta z^\beta$, the coefficient of $u^\gamma$ in $p(\partial)F$ is

$$\frac{1}{\gamma!} \sum_\beta p_\beta c_{\gamma+\beta}.$$

Thus the ordinary differential annihilator of $F$ is exactly the contraction annihilator of $C$. Let $A = \mathbb{C}[z_1, \dots, z_6]/\operatorname{Ann}(F)$ and $\lambda(p) = (p(\partial)F)(0)$. Its Gorenstein pairing $\lambda(pq)$ has rank $\dim A = 14$. On the standard Fock space $\mathcal{F}^2(\mathbb{C}^6)$, with $\|f\|_{\mathcal{F}^2}^2 = \sum_\alpha \alpha! |f_\alpha|^2$ for $f(z) = \sum_\alpha f_\alpha z^\alpha$, define $\Gamma(f,g) = \lambda(fg)$. This pairing uses only degrees through three, so it is a bounded Hankel form of rank fourteen.

Assume that sums of at most fourteen point-evaluation forms approach $\Gamma$ pointwise. Choose a quotient basis $p_1 = 1, p_2, \dots, p_{14}$ of $A$ with each $\deg p_i \le 3$. Its Gram matrix and the six mixed-moment matrices are

$$G_{ij} = \lambda(p_i p_j), \qquad (V_k)_{ij} = \lambda(p_j z_k p_i), \qquad M_k = G^{-1} V_k.$$

The Gorenstein pairing makes $G$ invertible. Every entry of $G$ and $V_k$ is a moment of degree at most seven, so pointwise convergence forces the approximating matrices $G_m, V_{k,m}$ to converge to $G, V_k$. Eventually $G_m$ is invertible. After combining repeated support points and deleting zero weights, write an approximant as $\lambda_m = \sum_{s=1}^{r_m} a_s \operatorname{ev}_{z_s}$. If $E_{is} = p_i(z_s)$ and $D = \operatorname{diag}(a_s)$, then $G_m = EDE^{\mathsf{T}}$. Its rank is fourteen, while $r_m \le 14$, so $r_m = 14$, all weights are nonzero, and $E$ is invertible. Moreover

$$M_{k,m} = G_m^{-1} V_{k,m} = E^{-\mathsf{T}} \operatorname{diag}(z_{1,k}, \dots, z_{14,k}) E^{\mathsf{T}}.$$

These matrices commute, and $p_i(M_{1,m}, \dots, M_{6,m})e_1 = e_i$ for every $i$. Passing to the limit preserves these polynomial equations.

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Any commuting tuple $M_1, \dots, M_6$ satisfying $p_i(M_1, \dots, M_6)e_1 = e_i$ defines a length-fourteen quotient: the map $q \mapsto q(M_1, \dots, M_6)e_1$ is onto, and its kernel is an ideal because the matrices commute. The equations specify a closed quotient-basis chart. The approximating evaluation tuples are reduced points of this chart, so their limit would make $A$ a flat limit of reduced length-fourteen schemes, that is, smoothable. This contradicts the choice of $C$. Only moments of degree at most seven were used, so a neighborhood defined by finitely many pointwise tests already excludes all sums of at most fourteen evaluations. This refutes the several-variable conjecture even in the pointwise topology and leaves the one-variable assertion unaffected. □

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